



INTRO

PREMISE

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Lifestyle-Based Depression
Risk Classification

SEMINAR PROJECT

PROJECT IDEA

Calculating risk of depression based on lifestyle factors—

- High risk/low-risk
- Sleep, diet, socialization, screen time, etc.

Dataset: [“Chinese University Students' Lifestyle and Physical-Mental Health Dataset \(2024\)” by Wei et al.](#)

The target output ***is not*** a clinical diagnosis!



DATASET

- Cutoff between low-risk and high-risk based on PHQ-9 medical concern cutoff
- 2529 low-risk individuals, 188 high risk individuals
- **Excluded:** University, province, PHQ (depression), GAD (anxiety), stress
- **Baseline:** Top 90% percentile of stress scores marked as high-risk

Over the last 2 weeks, how often have you been bothered by the following problems?		Not at all	Several Days	More than half the days	Nearly every day
1	Little interest or pleasure in doing things	0	1	2	3
2	Feeling down, depressed, or hopeless	0	1	2	3
3	Trouble falling asleep or sleeping too much	0	1	2	3
4	Feeling tired or having little energy	0	1	2	3
5	Poor appetite or overeating	0	1	2	3
6	Feeling bad about yourself- or that you are a failure or have let yourself or family down	0	1	2	3
7	Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8	Moving or speaking so slowly that other people could have noticed. Or the opposite-being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9	Thoughts that you would be better off dead, or of hurting yourself in some way	0	1	2	3
TOTAL SCORE (add the marked numbers):					

	count	proportion	
lower risk (0)	2529	0.930806	
higher risk (1)	188	0.069194	
Prepared shape: (2717, 83)			
Duplicate full rows: 0			
Total missing cells: 490			

THE APPROACH

MODELS

SPLIT

WEIGHTING

METRIC

01

- Logistic regression
- Gaussian Naive Bayes
- Radial Basis Function kernel (RBF SVC)

Low-risk represented as 0, high-risk as 1.

02

The dataset was split using the 70/15/15 model. 70% used for training, 15% for validation, and 15% as the test set. Randomly sorted. Preserves rate of high risk individuals in whole population.

03

False positives (marking low-risk as high-risk) weighted heavier than missed opportunities (marking low-risk as high-risk) due to category imbalance.

04

1. Recall
2. Precision
3. F-score (harmonic mean of precision and recall)
4. PR-AUC (plotting area under curve)
5. ROC-AUC (used for binary classification)

...

rows higher_risk_n higher_risk_pct



train

1901

132

6.943714



validation

408

28

6.862745

test

408

28

6.862745

MODEL COMPARISON

...	model	params	threshold	recall	precision	f1	flag rate	pr_auc	roc_auc
0	gaussian_nb	{'model__var_smoothing': 1e-07}	0.99	0.928571	0.206349	0.337662	0.308824	0.294202	0.872603
1	gaussian_nb	{'model__var_smoothing': 1e-08}	0.99	0.928571	0.181818	0.304094	0.350490	0.261938	0.863111
2	gaussian_nb	{'model__var_smoothing': 1e-09}	0.99	0.928571	0.160494	0.273684	0.397059	0.250129	0.858271
3	gaussian_nb	{'model__var_smoothing': 1e-10}	0.99	0.928571	0.137566	0.239631	0.463235	0.243858	0.855310
4	rbf_svc	{'model__C': 0.5, 'model__gamma': 0.01}	0.10	0.892857	0.265957	0.409836	0.230392	0.432942	0.908365
5	logistic_regression	{'model__C': 0.1}	0.31	0.892857	0.240385	0.378788	0.254902	0.469471	0.898872
6	rbf_svc	{'model__C': 1.0, 'model__gamma': 0.01}	0.02	0.857143	0.279070	0.421053	0.210784	0.433127	0.903947
7	logistic_regression	{'model__C': 1.0}	0.18	0.821429	0.205357	0.328571	0.274510	0.421128	0.880451
8	logistic_regression	{'model__C': 10.0}	0.26	0.714286	0.224719	0.341880	0.218137	0.437005	0.865789
9	rbf_svc	{'model__C': 0.5, 'model__gamma': 'scale'}	0.01	0.678571	0.287879	0.404255	0.161765	0.436173	0.905451
10	rbf_svc	{'model__C': 1.0, 'model__gamma': 'scale'}	0.01	0.321429	0.600000	0.418605	0.036765	0.431386	0.884117
11	rbf_svc	{'model__C': 5.0, 'model__gamma': 0.01}	0.01	0.285714	0.500000	0.363636	0.039216	0.410750	0.873214
12	rbf_svc	{'model__C': 5.0, 'model__gamma': 'scale'}	0.14	0.142857	0.800000	0.242424	0.012255	0.394669	0.817011
13	rbf_svc	{'model__C': 0.5, 'model__gamma': 0.1}	0.01	0.000000	0.000000	0.000000	0.000000	0.361067	0.885338
14	rbf_svc	{'model__C': 1.0, 'model__gamma': 0.1}	0.01	0.000000	0.000000	0.000000	0.000000	0.361015	0.885244
15	rbf_svc	{'model__C': 5.0, 'model__gamma': 0.1}	0.01	0.000000	0.000000	0.000000	0.000000	0.361015	0.885244

Best model: Gaussian Naive Bayes

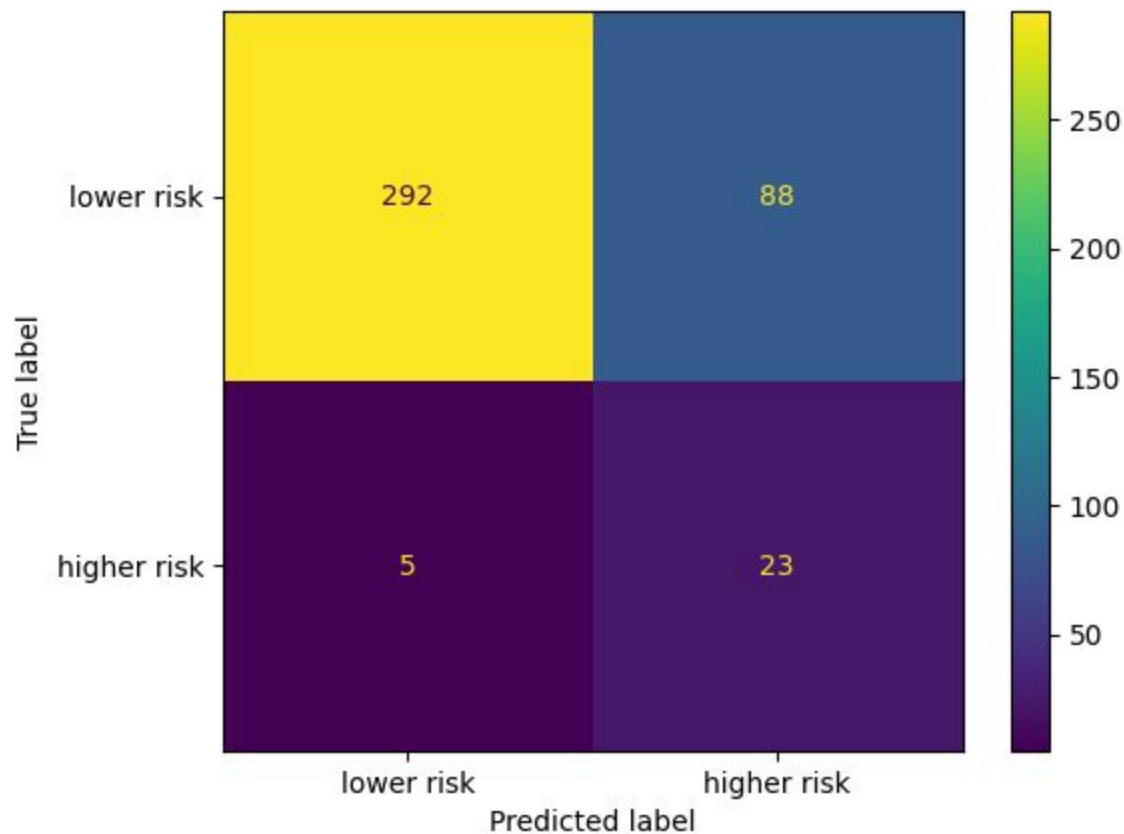
- High recall, low precision

flag_rate: % of population flagged as high-risk

Higher recall compared to base rate

Training-set 90th-percentile Psy Stress cutoff: 22.000									
	training_stress_cutoff	training_stress_median	recall	precision	f1	flag_rate	pr_auc	roc_auc	
stress_90th_percentile_rule	22.0	16.0	0.571429	0.410256	0.477612	0.095588	0.538521	0.902491	

Final test confusion matrix



THOUGHTS

GNB

Assumes that features are conditionally independent—which is not true for this dataset (e.g. screen time and sleep quality)! I was surprised by how well it worked.

Generalizability

All the data was collected from university students in China. Especially when weighing different factors (like in logistic regression), I think it could be relevant to consider that different populations may have different standards for a healthy lifestyle (e.g. older people may need more sleep).

Linearity

Logistic regression assumes a linear relationship between variables, while SVC is beneficial for non-linear relationships. What is the true relationship between the different features?

Dataset Size

Another important factor when comparing algorithms—GNB does better on smaller datasets.

Thank You

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2026.07.16

Any questions or comments?

Repo link:

<https://github.com/yokunerukosb/elt/summer-2026-seminar-paper>